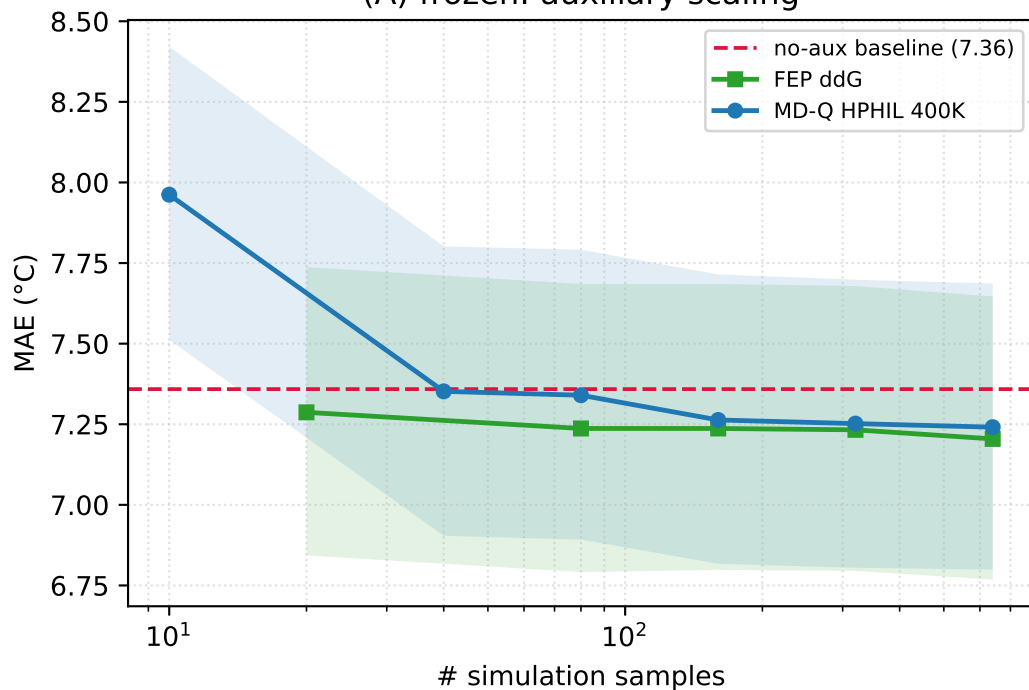
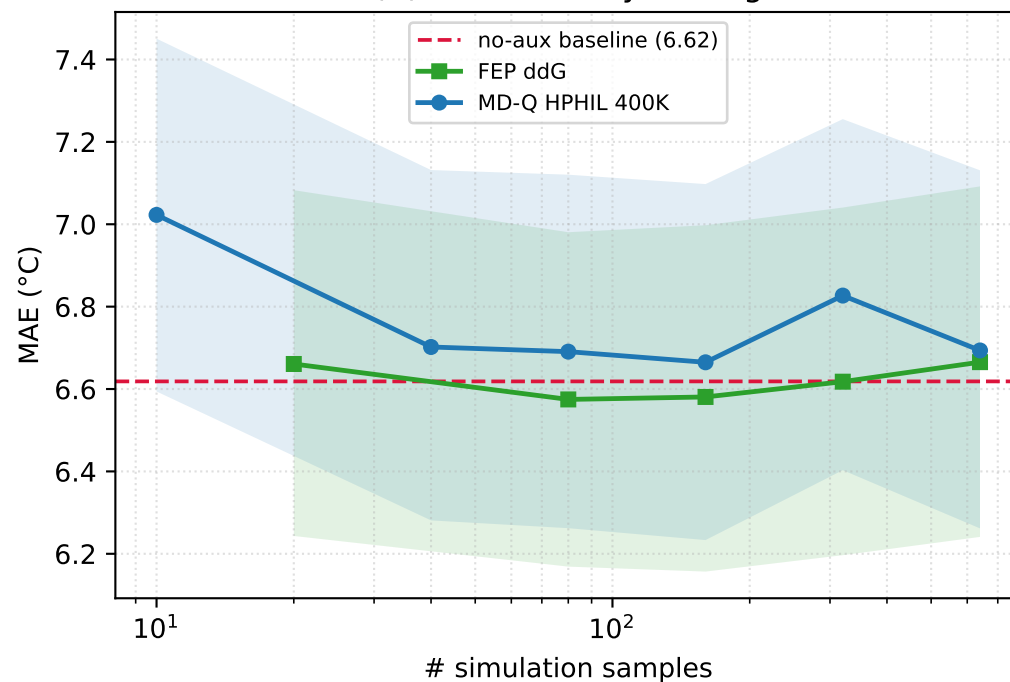


Same-pipeline comparison: conventional MD-Q $\approx$ physics ddG (FEP); hot encoder saturates

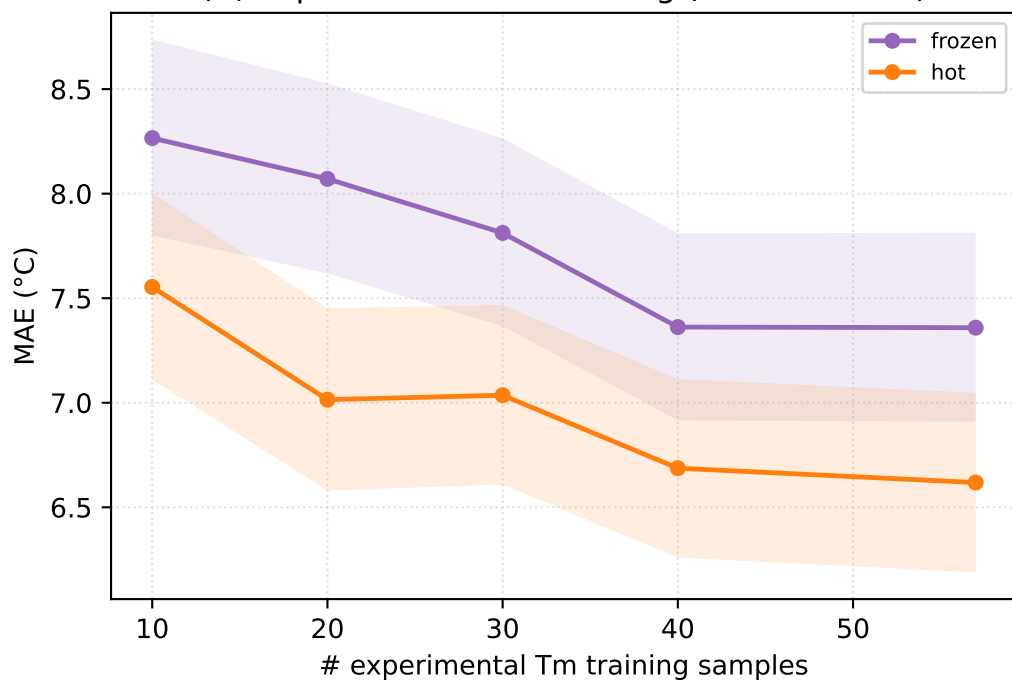
(A) frozen: auxiliary scaling



(B) hot: auxiliary scaling



(C) experimental-data scaling (no simulation)



(D) best MAE by condition

